Failure Analysis Report

FAIL-0105 — Resource Exhaustion

1. Failure Summary

Failure ID	FAIL-0105
Failure Type	Resource Exhaustion
Severity	Critical
Component	Bluetooth Module
Application	APP-003 — GalaxySearch
Build Version	6.1.7
Timestamp	2023-02-12 01:05:20
Environment	ENV-001 — Mobile Lab
OS Version	Android 14
Network	WiFi-6
Resolution	Resolved

2. Test Case Reference

Test Case ID	TC-1846
Title	Confirm that firmware over-the-air update triggers the expected notification whe...
Execution ID	EXEC-00458
Test Run	RUN-006 — Regression Cycle 6 — Build 5.2.1
Requirement	REQ-0393 — Functional — Accessibility Support

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete search query execution within the defined SLA.
Observed Behaviour	The system exhibited resource exhaustion when two concurrent requests are made simultaneously.

4. Error Details

Error Message:

```
OOMError: heap exhausted at 695MB/2048MB during sync_batch.
```

Stack Trace / Log Excerpt:

```
com.samsung.browser.NetworkStack.execute(line 359)
```

Log Reference	LOG-52807

5. Executed Steps at Time of Failure

SYNTHETIC DATA — No real Samsung confidential information

1	Navigate to Settings > Account and tap 'Verify Identity'.	A confirmation notification appears on the paired device.
2	Verify that the UI displays 'Error: retry'.	A confirmation notification appears on the paired device.
3	Capture a screenshot and attach as evidence.	Server returns HTTP 200 with a valid JSON payload.
4	Verify that the UI displays 'Error: retry'.	The UI transitions to the next screen within 3 seconds.

6. Root Cause Analysis

Firmware update binary not validated against checksum before flashing; partial write leaves device in boot loop.

8. Historical Similar Failures

HIST-0104	FAIL-0105	Resource Exhaustion	3.0.2	2023-03-24	3	Resolved
HIST-0200	FAIL-0201	Resource Exhaustion	6.2.5	2021-01-24	13	Resolved
HIST-0690	FAIL-0691	Resource Exhaustion	3.0.2	2023-01-04	2	Open

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

SYNTHETIC DATA — No real Samsung confidential information